5 **(a)** Any three of the following: B3

- wave profile of stationary waves does not advance
- amplitude varies from zero at node to maximum at antinode.
- particles within the **internodal segment** oscillate in phase while particles in **adjacent internodal segments** oscillate anti-phase
- particles oscillate in SHM with same frequency except at the nodes
- no wavelength
- distance between adjacent nodes (or antinodes) equal half the wavelength of the waves that form the stationary wave

(b) waves from source reflected by reflective casing B1

incident and reflected wave (with same amplitude) overlap and superpose B1

(c) wavelength = $5.8 \times 2 = 11.6$ cm C1

speed = $2.45 \times 10^9 \times 0.116 = 2.8 \times 10^8$ or 2.84×10^8 m s^{-1} A1

(d) wavelength of microwave (11.6 cm) > 2 mm, wavelength of visible light (~10^{-7} m) < 2 mm B1